



MARMARA
ÜNİVERSİTESİ

FACE AND IDENTITY VERIFICATION USING CLOUD-BASED SAAS

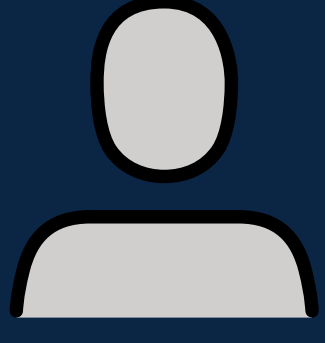


A MULTI-TENANT BIOMETRIC IDENTITY PLATFORM — PRODUCTION-DEPLOYED

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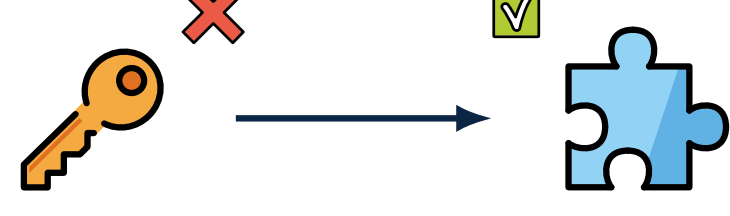
Register once. Verify everywhere. Every time.



Ten authentication factors, one hosted login—protected by a liveness check a recorded video cannot fake.

What FIVUCSAS solves

Passwords leak,
SIM-OTPs get swapped



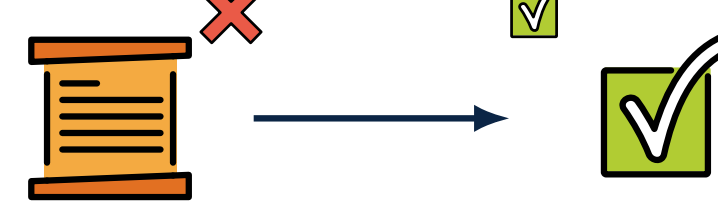
10 composable factors
in one MFA flow

Photos, clips & deepfakes
replay a face



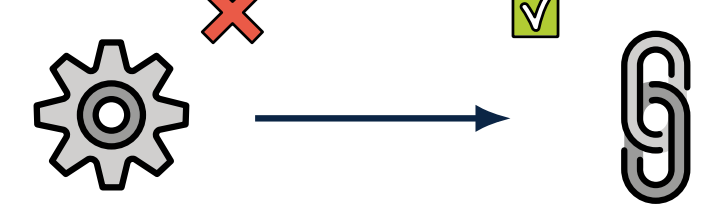
Active randomised
liveness challenge

GDPR & KVKK bolted on
as an afterthought



Consent, export &
erasure built in

Re-integrating auth
for every new app



One hosted OIDC
for every platform

Questions FIVUCSAS answers

✓ **Is this a real person
— or a replay?**
Active liveness—0% replay false-accept (n=120)

✓ **Can one identity work
across many apps?**
Hosted OAuth2 / OIDC, tenant-scoped—redirect
once

✓ **Is my biometric
data safe?**
Facenet512 embeddings encrypted at rest—AES-128
Fernet

✓ **Can I delete
my data?**
GDPR Art. 17 erasure + scheduled purge—KVKK
aligned

✓ **Do you verify real-world
identity — not just login?**
9-step pipeline: document + NFC chip + face +
liveness

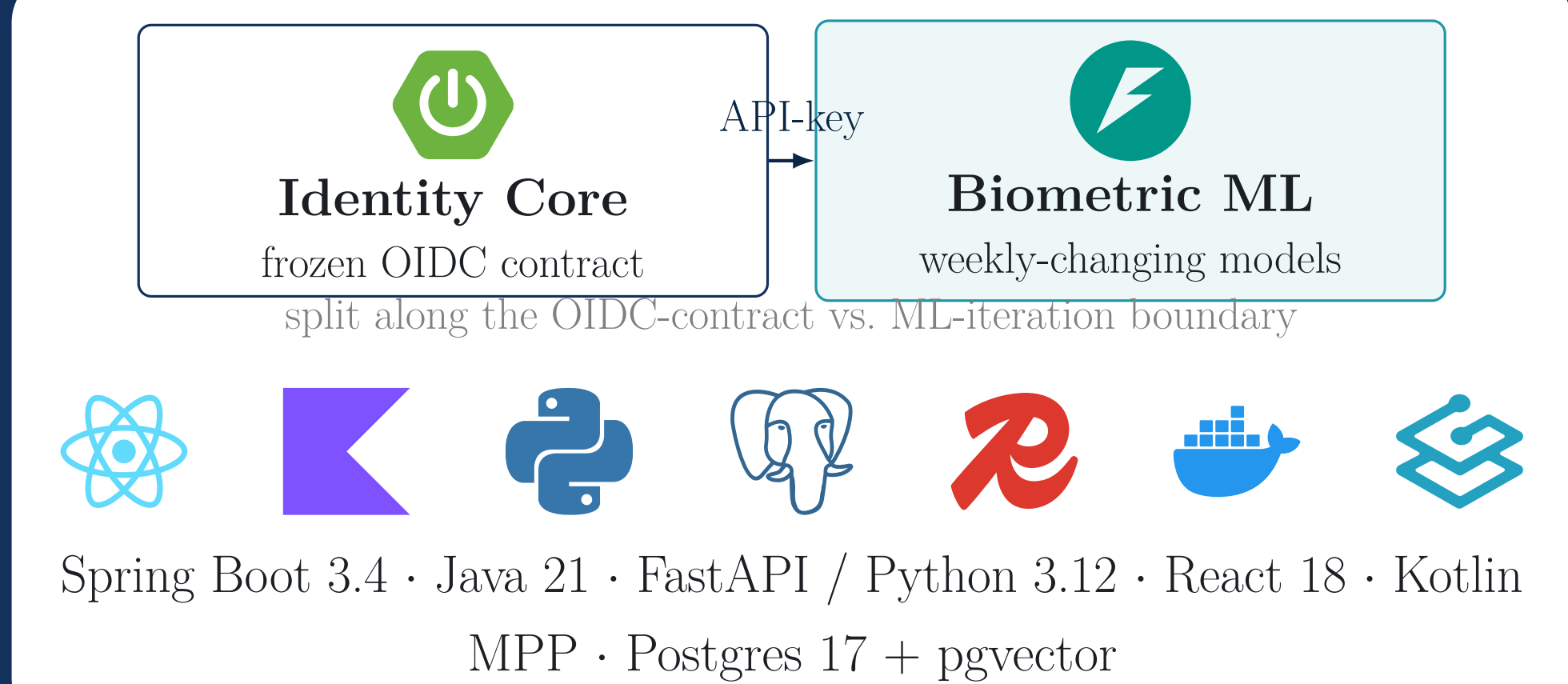
Top 10 features

- 1 **Ten composable auth factors**
password · OTP · TOTP · QR · face · voice · NFC ·
WebAuthn
- 2 **Active-liveness anti-spoofing**
random facial-action challenge + passive MiniFASNet
- 3 **Hosted OAuth2 / OIDC + PKCE**
redirective login, JWKS, discovery, RFC 8252 loop-
back
- 4 **Multi-tenant, tenant-controlled flows**
each tenant composes its own MFA flow, no backend
code
- 5 **RBAC & fine-grained permissions**
roles, user-roles and permissions, row-level security
- 6 **9-step identity-verification pipeline**
document scan → NFC chip → face match → live-
ness
- 7 **7+ industry verification templates**
Banking KYC, Healthcare, Gov e-KYC, Fintech...
- 8 **Embedding encryption at rest**
Fernet AES-128 wrap before pgvector write
- 9 **GDPR / KVKK compliance built in**
consent, data export, right-to-erasure + purge
- 10 **Every platform**
web · Android · iOS · desktop · CLI

By the numbers

1820+	< 1 s
automated tests	face verification (P95)
0 %	10
replay false-accept	auth factors
9	7+
verification steps	industry templates
8	717
live production services	commits shipped

Architecture at a glance



See it live



scan to try it

fivucasas.com
verify.fivucasas.com
app.fivucasas.com
api.fivucasas.com